

1. Full Forms and One-Line Use

Term	Full Form	Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Used for secure communication over the web using encryption.
FTP	File Transfer Protocol	Used to transfer files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send emails from one mail server to another.

2. Difference Between IPv4 and IPv6 (2 Lines)

- **IPv4** uses 32-bit addresses and supports about 4.3 billion unique addresses (e.g., 192.168.1.1).
 - **IPv6** uses 128-bit addresses and supports a much larger number of unique addresses (e.g., 2001:db8::1).
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3. nslookup google.com

I cannot run commands on your computer. You can run:

```
nslookup google.com
```

Example output may show an IP address such as:

```
142.250.193.14
```

Note: The IP address may vary depending on your location and Google's servers.

4. Meaning of HTTP Status Codes

Status Code	Meaning
200	OK – The request was successful.
301	Moved Permanently – The page has been permanently redirected to a new URL.
403	Forbidden – Access to the requested resource is denied.
404	Not Found – The requested page or resource does not exist.
500	Internal Server Error – The server encountered an unexpected error.

5. DNS Hierarchy with www.google.com

DNS (Domain Name System) is organized in a hierarchy:

1. **Root (.)** – The top level of the DNS hierarchy.
2. **TLD (Top-Level Domain)** – com
3. **Domain Name** – google
4. **Subdomain** – www
5. Root (.)

```
|
com ← TLD
|
google ← Domain
|
www ← Subdomain
```